PersistRank: Duration-Aware Alert Prioritization for E-Commerce Microservice Monitoring*

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Abstract

In e-commerce microservices, large volumes of anomalies are generated simultaneously, requiring prioritization ranking for systematic resolution of anomalies. Ignoring persistent but seemingly less critical anomaly flags makes the whole system architecture prone to collapse during service delivery. This work proposes a novel approach,PersistRank, incorporating persistence score which includes the duration of the alert as another vital factor for ranking anomaly alerts. This approach enables the ranking method to count time windows showing the duration of existing anomalous services. The evaluation is done using the RS-anomic dataset and PersistRank outperforms the severity-criticality baseline by 7.2 percentage points in P@3. An ablation study confirms that persistence has been treated as an independent and complementary dimension in this approach.

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